Dentinogenesis Tutorial

Dr. Ume Hani

IDENTIFY, STUDY, AND PASTE THE DIAGRAMS

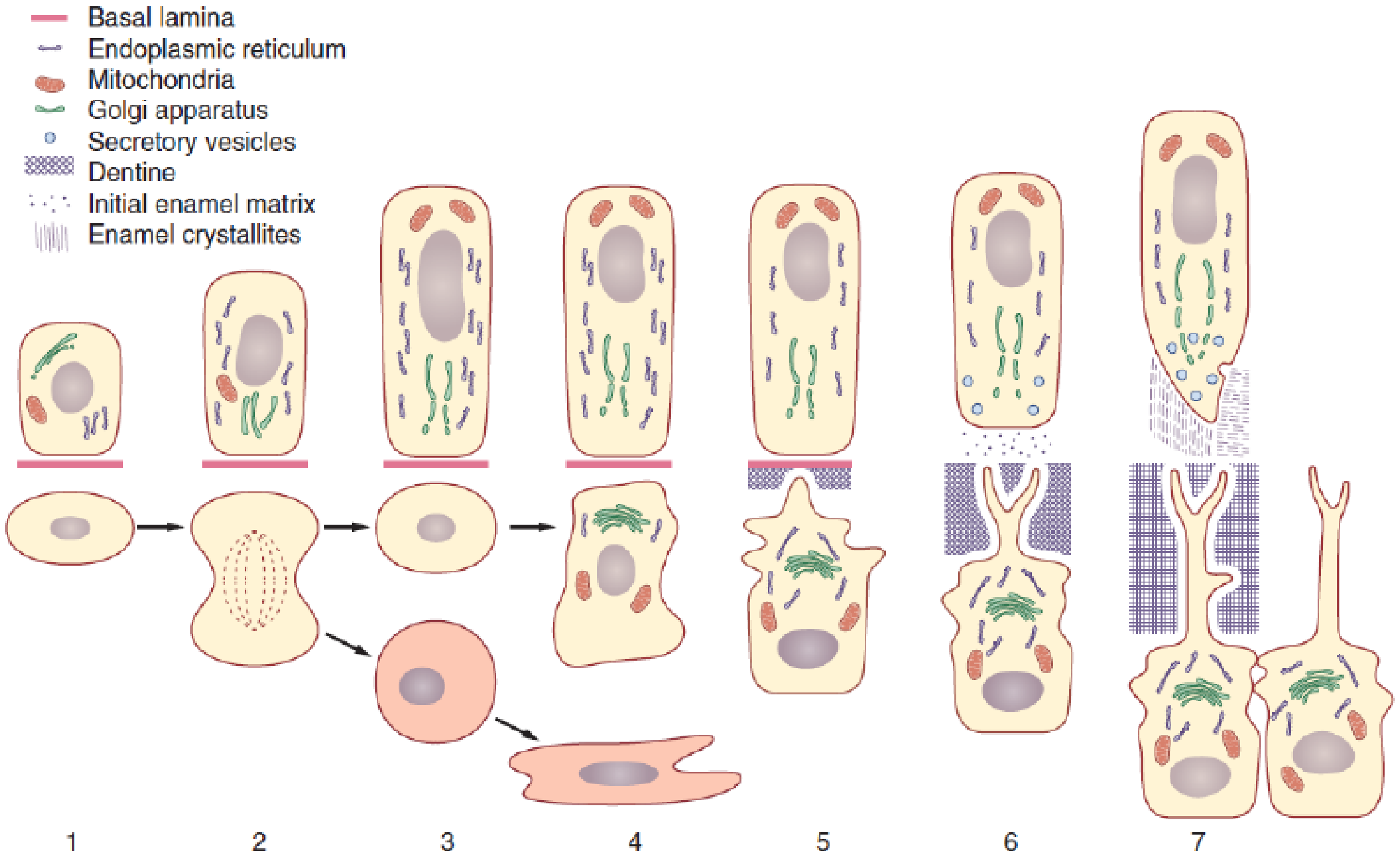

Fig. 23.1 Life cycle of the odontoblast (lower cell line) related to that of the ameloblast (upper cell line). 1 = Ameloblast begins to differentiate first. 2 = Peripheral ectomesenchymal cells divide, with some daughter cells migrating below the odontoblast layer. 3 = Acting on a signal from the ameloblast, the preodontoblasts begin to differentiate. 4 = Synthetic organelles increase in size and number, especially the Golgi apparatus and rough endoplasmic reticulum. 5 = Nucleus moves basally as the cell becomes polarised. A number of odontoblast processes begin to form. One odontoblast process becomes enlarged and begins to secrete matrix. 6 = The odontoblast retreats as matrix is laid down, leaving behind a single main process. Once a narrow layer of matrix is laid down mineralisation commences. 7 = Once the first layer of dentine is laid down the differentiated ameloblast begins to deposit matrix.

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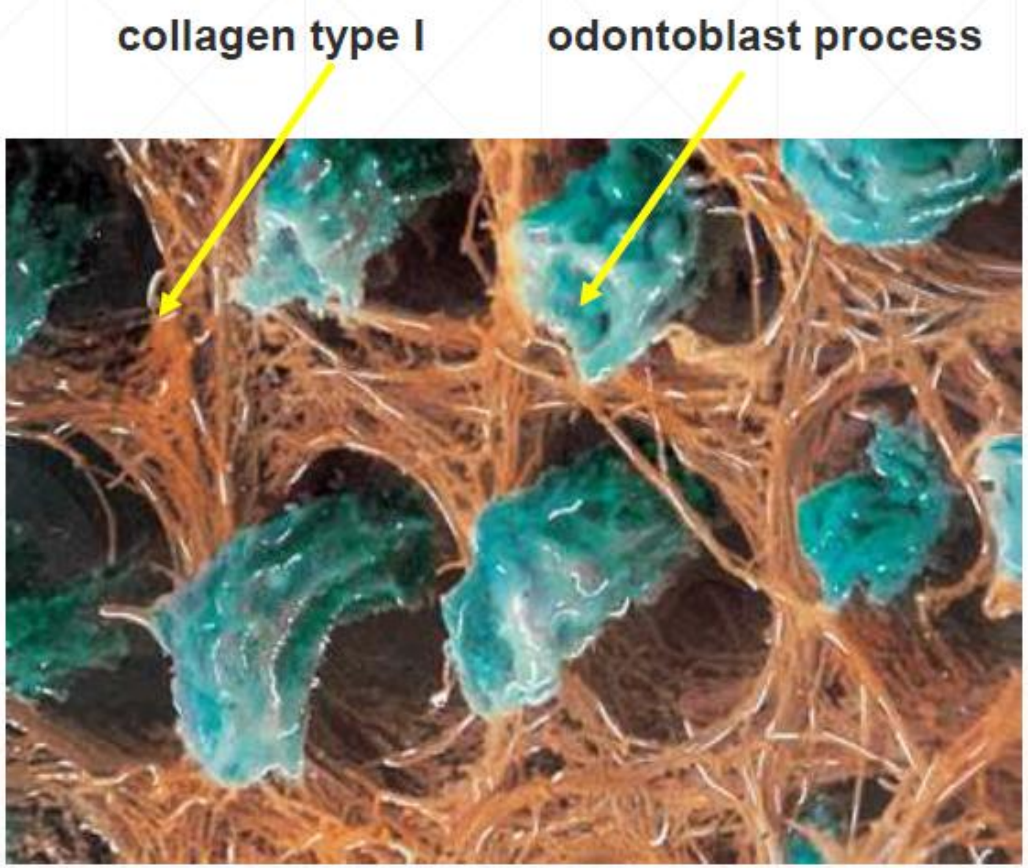
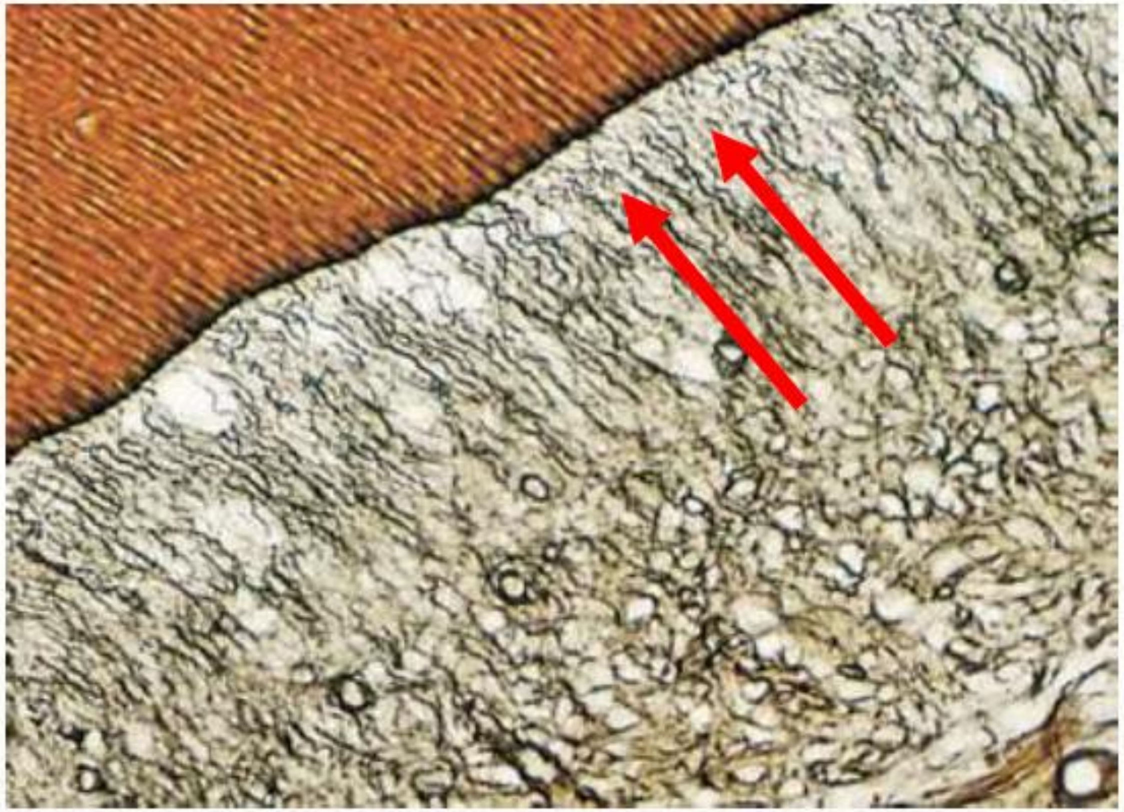


SEM showing the surface of predentine. The collagen fibrils form an interlacing network perpendicular to the odontoblast process.



Micrograph of a demineralised section showing 'corkscrew' fibres (von Korff's) shown by arrow.

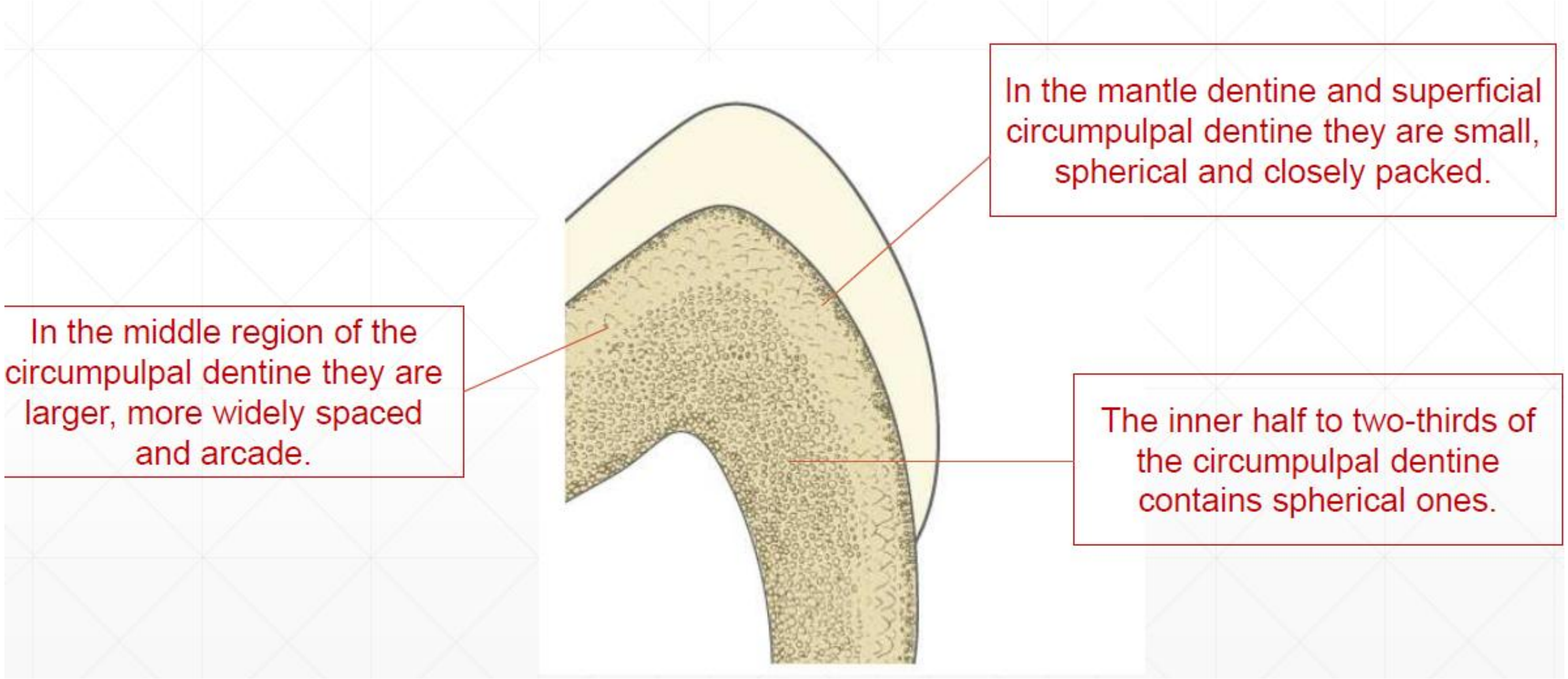


Fig. 23.19 The distribution of calcospherites in dentine. Most calcospherites are spherical. However, many have an arcade shape, such that the round apex of the arcade is directed towards the outer surface of the dentine and the opening is directed towards the pulp.

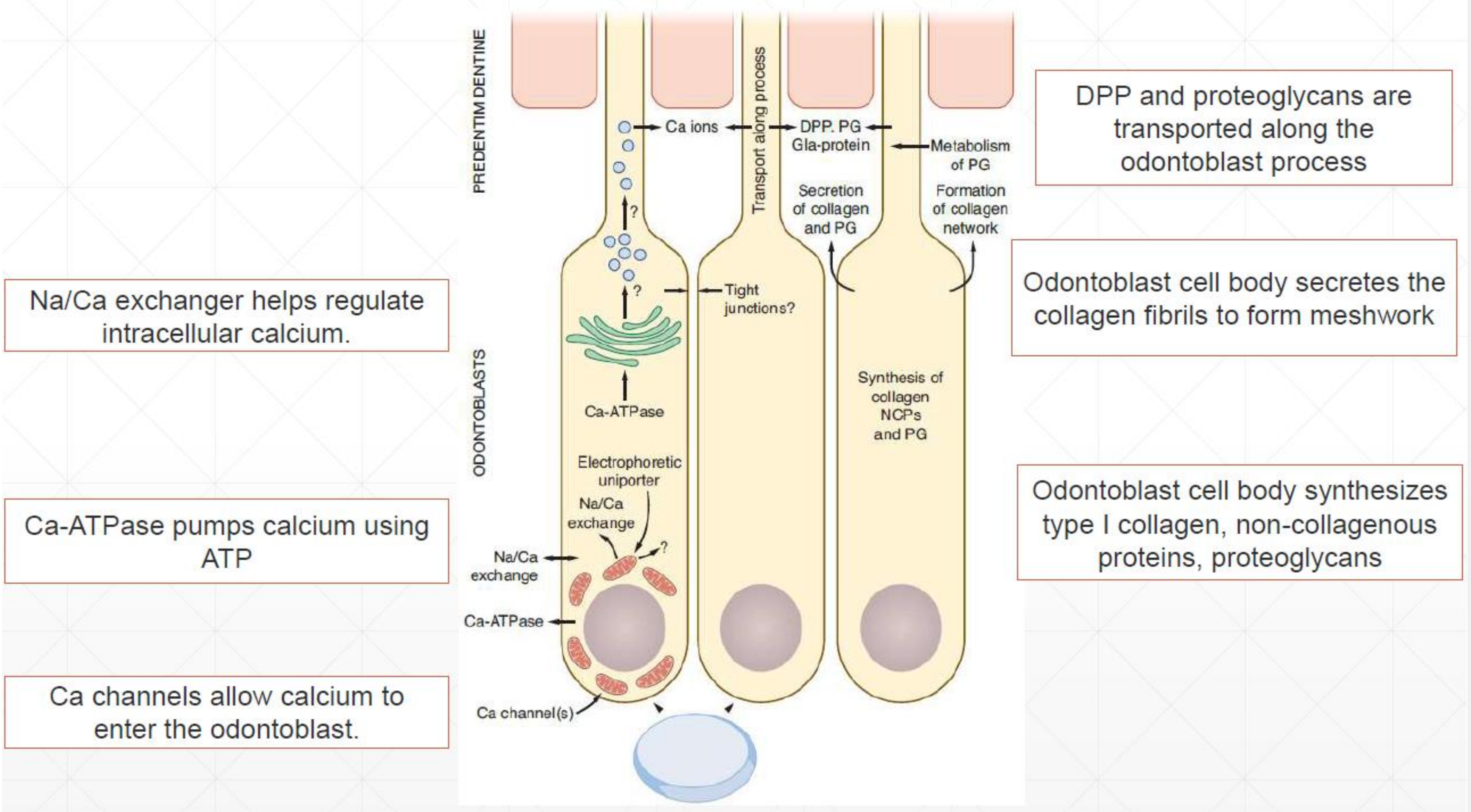


Fig. 23.10. Schematic diagram of fully differentiated active odontoblasts showing different calcium ion transport mechanisms (left side of figure) as well as putative transport routes for the dentine matrix macromolecules (right side of figure).

Fig. 23.25 A patient with dentinogenesis imperfecta. The teeth are malformed and are dark because of the absence of pulp chambers and the consequent loss of translucency. The enamel is normal but flakes off as it is only weakly attached to the dentine.